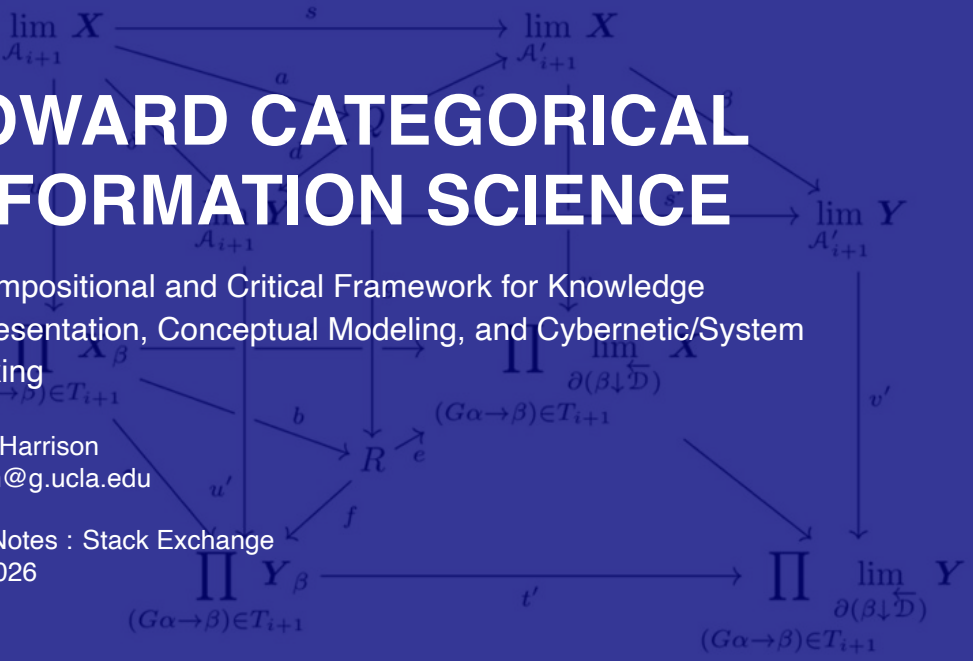
TOWARD CATEGORICAL INFORMATION SCIENCE

A Compositional and Critical Framework for Knowledge Representation, Conceptual Modeling, and Cybernetic/System Thinking

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Field Notes : Stack Exchange
Dec 2026



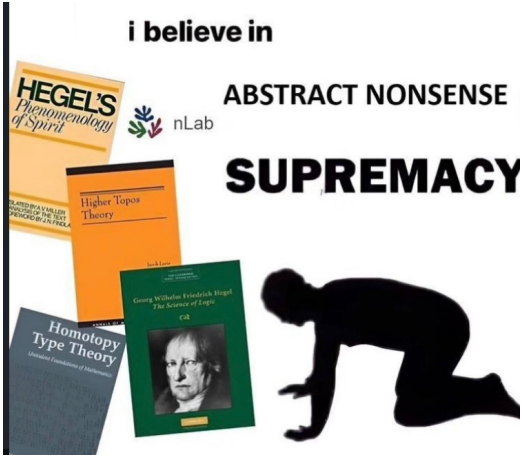
Overview

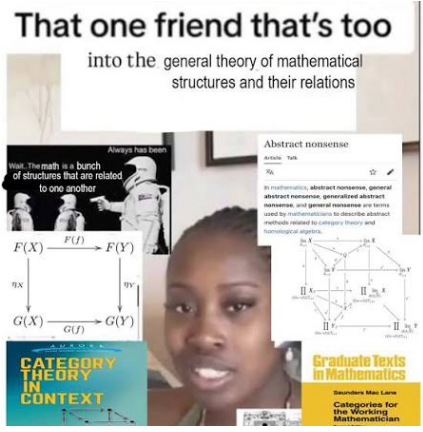
1. Motivation
2. Philosophical Preliminaries
3. The Categorical Paradigm
4. What is Category Theory?
5. Applied Category Theory
6. Knowledge Representation using Ologs
7. Functorial Data Migration and Categorical Databases
8. Categorical cybernetics
9. Socialism???

“

To us power is, first of all, the ability to define phenomena, and secondly the ability to make these phenomena act in a desired manner.

— Huey Newton


i believe in
HEGEL'S Phenomenology of Spirit
nLab
ABSTRACT NONSENSE
SUPREMACY
Higher Topos Theory
Georg Wilhelm Friedrich Hegel The Science of Logic
Homotopy Type Theory


That one friend that's too
into the general theory of mathematical structures and their relations
Always has been
Wait. The math is a bunch of structures that are related to one another
Abstract nonsense
In mathematics, abstract nonsense, general abstract nonsense, generalized abstract nonsense, and general nonsense are terms used by mathematicians to describe abstract methods related to category theory and homological algebra.
CATEGORY THEORY IN CONTEXT
Graduate Texts in Mathematics
Saunders Mac Lane
Categories for the Working Mathematician

“

The language of categories is affectionately known as “abstract nonsense,” so named by Norman Steenrod. This term is essentially accurate and not necessarily derogatory: categories refer to “nonsense” in the sense that they are all about the “structure,” and not about the “meaning,” of what they represent.

— Paolo Aluffi, *Algebra: Chapter 0*

MOTIVATION

Motivation

Categorical Information Science is my attempt to think at the intersection of philosophy, information science, and applied category theory at three abstract levels

- level of concept
- level of problem
- level of system

In this presentation I present an assemblage of ideas, examples and provocations for thinking about ways that Applied category theory can inform theorizing, modeling, and scientific practice.

The Aristotilian and Set theoretic Legacy of Information Science While we don't often focus explicitly on mathematics and logic within information science, there is a hidden legacy which informs much of thinking and practice within IS

PHILOSOPHICAL PRELIMINARIES

The reading list

- Sylvia Wynter
- Yuk Hui
- Hegel
- Robert Brandom
- Celia Lury
- Lawvere & Schanuel
- Eugenia Cheng
- Alain Badiou
- Emily Reihl
- David Spivak
- Brenden Fong
- ...

THE CATEGORICAL PARADIGM

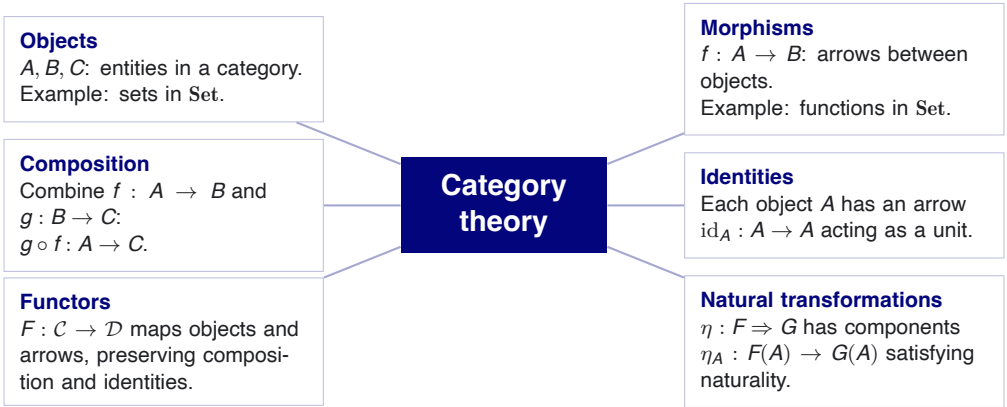


Mathematical objects are determined by–and understood by–the network of relationships they enjoy with all the other objects of their species.

— Barry Mazur

WHAT IS CATEGORY THEORY?

Category theory: a map of the basic concepts



Category laws $h \circ (g \circ f) = (h \circ g) \circ f \qquad \mathrm{id}_B \circ f = f = f \circ \mathrm{id}_A$

Objects and arrows live within categories; functors connect categories; natural transformations connect parallel functors.

Common mathematical domains and the categorical viewpoint

Category	Objects	Morphisms
Set	sets	functions
Group	groups	group homomorphisms
Coproduct $A + B$	Injections from A and B	Disjoint union with source tags

APPLIED CATEGORY THEORY

Themes

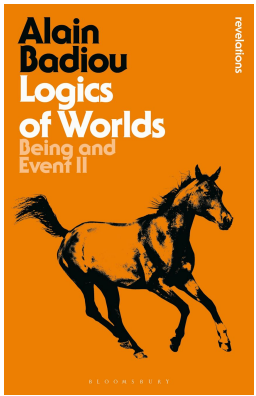
- category theory as a model of models
- functorial semantics
- compositionality

KNOWLEDGE REPRESENTATION USING OLOGS

FUNCTORIAL DATA MIGRATION AND CATEGORICAL DATABASES

CATEGORICAL CYBERNETICS

SOCIALISM???



"The human animal is the being of the thousand logics" (Badiou, 2009, pg 69).

Logic of Worlds by Alain Badiou

In his second entry in the being and event series Badiou complements his set theoretic ontology of pure being with a systematic philosophy of appearance, phenomenology and worlds using category theory.

There is no single universal logic that governs appearance, instead there are many worlds governed by logics

Logic

What are the entities and distinctions in the domain?

- concepts
- relations
- categories

Appearance

Worlds are formalized as categories where objects appear according to a transcendental, an algebra that assigns degrees of exisstence to objects and their relations.

- goals
- constraints
- evidence

Objective Phenomenology and the Logic of Capital

Phenomenology typically focuses on how things appear to the consciousness of a human subject.

Objective Phenomenology Suspends the subject and focuses on how objects and social forms appear to each other in the logic of a specific world.

Marx as Objective Phenomenologist

Marx critique of commodity fetishism points us to how systemic relations between things are reduced human perceptions of seeminly innate properties

Gabriel Tupinamba, Yuan Yao

APPENDIX

Two-column comparison

Concept space

What are the entities and distinctions in the domain?

- concepts
- relations
- categories

Problem space

What question or difficulty are we trying to address?

- goals
- constraints
- evidence

Three-step process

01

Observe
Identify the domain and its questions.

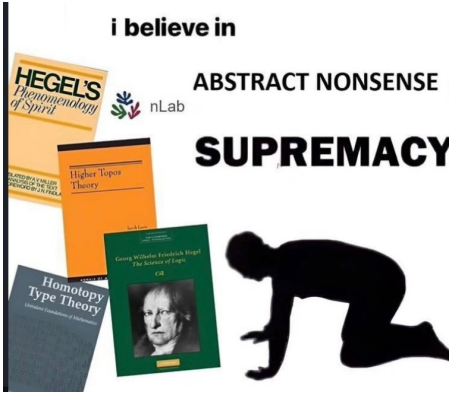
02

Model
Represent concepts and relationships.

03

Reflect
Test what the model makes visible.

Image with a takeaway



A visual can do more than decorate.

Use the image to establish a setting, then connect it to one clear claim.

Image caption or source goes here.

Observation

A model groups several distinct cases under the same category.

- What is included?
- What is left out?

→

Interpretation

Classification choices shape what can be compared, counted, or acted upon.

- State the implication.
- Name the limitation.

ONE TAKEAWAY

Models do not only describe a world. They help determine what we can ask of it.

Functor

A functor $F : \mathcal{C} \to \mathcal{D}$ maps objects and morphisms while preserving identities and composition:

$$F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$$

$$\boxed{F(g \circ f) = F(g) \circ F(f)}$$

What this means

- Objects become objects.
- Arrows become arrows.
- Composing before or after translation gives the same result.

Math: a step-by-step derivation

Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{E}$ be functors, with composable arrows $f : X \to Y$ and $g : Y \to Z$.

$$\begin{aligned}(G \circ F)(g \circ f) &= G\big(F(g \circ f)\big) \\ &= G\big(F(g) \circ F(f)\big) \\ &= G(F(g)) \circ G(F(f)) \\ &= (G \circ F)(g) \circ (G \circ F)(f).\end{aligned}$$

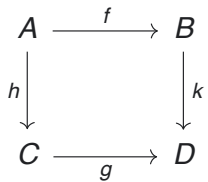
Takeaway: the composite preserves composition. Identity preservation follows similarly, so $G \circ F$ is a functor.

Table: comparing categorical constructions

Construction	**Data**	**Interpretation in sets**
Product $A \times B$	Projections to A and B	All pairs (a, b)
Pullback $A \times_C B$	Maps $f : A \to C$, $g : B \to C$	Pairs with $f(a) = g(b)$
Coproduct $A + B$	Injections from A and B	Disjoint union with source tags

These constructions are characterized by universal properties, not just by their set-theoretic realizations.

Diagram: a commutative square



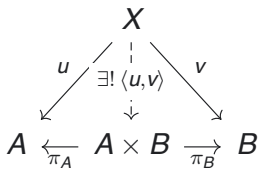
Two routes, one result

Starting at A and ending at D, either path gives the same morphism:

$$\boxed{k \circ f = g \circ h}$$

Commutativity is a condition on the arrows, not just a feature of how they are drawn.

Diagram: the universal property of a product



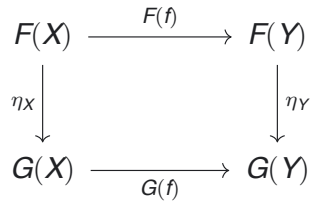
For every pair $u : X \to A$ and $v : X \to B$, there is a **unique** mediating arrow such that

$$\pi_A \circ \langle u, v \rangle = u,$$
$$\pi_B \circ \langle u, v \rangle = v.$$

In sets: $\langle u, v \rangle(x) = (u(x), v(x))$.

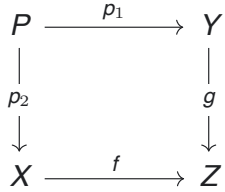
Diagram: naturality as compatibility

Let $F, G : \mathcal{C} \to \mathcal{D}$ be functors. A natural transformation $\eta : F \Rightarrow G$ assigns an arrow $\eta_X : F(X) \to G(X)$ to each object X.



For every $f : X \to Y$, the square commutes:

$$G(f) \circ \eta_X = \eta_Y \circ F(f).$$


P
p1
Y
p2
g
X
f
Z